PARAMETRIZED LOCAL YEO THEOREM FOR LOCALLY COLORED PARTIAL GRAPHS

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Allowed standard prerequisites (editorial).

- Finite partial graph paths and elementary ordering facts. Author §2 and §3 definitions; Lemmas4/5 are stated without separate proof in the publication and retained unchanged.

2 Graphs and Cusps

2.1 Partial Undirected Graphs and Paths

As we take interest in proof nets and Yeo's theorem in this paper, we study undirected paths in finite undirected partial multigraphs. We recall here quickly some basic notions from graph theory, for more details we refer the reader to [2].

A (finite undirected multi) partial graph (without loop) is a triple $(\mathcal{V}, \mathcal{E}, \psi)$ where $\mathcal{V}$ (vertices) and $\mathcal{E}$ (edges) are finite sets and ψ (the incidence function) associates to each edge a set of at most two vertices. An edge e is total when $\psi(e)$ is of cardinal two, and a total graph (or simply a graph) is one whose edges are total. Many notions lift immediately from total graphs to partial graphs, e.g. isomorphisms – that we denote $\simeq$. An edge e is incident to a vertex v if $v \in \psi(e)$, in which case v is an endpoint of e.

A path p is a finite alternating sequence of vertices and edges $(v_0, e_1, v_1, e_2, v_2, \dots, e_n, v_n)$ such that for all $i \in \{1; \dots; n\}$, the endpoints of e_i are exactly v_{i-1} and v_i (which are distinct). A path always has at least one vertex, but it can have no edge and be reduced to a single vertex (v_0), in which case it is called an empty path. With the notation above, v_0 is the source of p, v_n is its target and both make the endpoints of p. Since a given vertex may occur more than once in a path, we may have to talk about occurrences of vertices in a path to distinguish these equal values. We use the following notations:

■ the concatenation of two paths $p_1 = (v_0, e_1, \dots, e_k, v_k)$ and $p_2 = (v_k, e_{k+1}, \dots, e_n, v_n)$ is the path $p_1 \cdot p_2 = (v_0, e_1, \dots, e_k, v_k, e_{k+1}, \dots, e_n, v_n)$;

■ the reverse of a path $p = (v_0, e_1, v_1, \dots, e_k, v_k)$ is $\overline{p} = (v_k, e_k, v_{k-1}, \dots, e_1, v_0)$;

■ if v and u are two (occurrences of) vertices of a path p, with v occurring before u, $p_{(v,u)}$ is the unique sub-path (i.e. sub-sequence that is a path) of p with source v and target u.

A path is simple if its edges are pairwise distinct and its vertices are pairwise distinct except possibly its endpoints which may be equal. A path is closed if it has equal endpoints, otherwise it is open. A cycle is a non-empty simple closed path.

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Given a partial graph $G = (\mathcal{V}, \mathcal{E}, \psi)$, a sub-graph of G is a partial graph $G' = (\mathcal{V}', \mathcal{E}', \psi')$ such that $\mathcal{V}' \subseteq \mathcal{V}$, $\mathcal{E}' \subseteq \mathcal{E}$ and ψ' is the restriction of ψ to $\mathcal{E}'$ in its domain and sets of $\mathcal{V}'$ in its codomain. Connectedness is not immediate to define in partial graphs because paths go from vertices to vertices. Two vertices v and u are connected when there exists a path with endpoints v and u. Two edges are connected if they are incident to two connected vertices. An edge e and a vertex v are connected if e is incident to a vertex connected to v. A partial graph G is connected when any two different vertices or edges of G are connected. A connected component is a non-empty connected sub-graph maximal for the inclusion.

2.2 Local Coloring and Cusps

Let G be a (partial) graph. A local coloring of G is a function c taking as arguments an edge e and one of its endpoints v. By convention, the elements $\mathrm{c}(e, v)$ are called colors. The intuition is that given an edge e and one of its endpoints v, $\mathrm{c}(e, v)$ is the color of e according to v. A local coloring can also be seen as a coloring of half-edges, i.e. $\mathrm{c}(e, v)$ is the color of the half of e near v. When drawing a graph, we therefore represent $\mathrm{c}(e, v)$ by coloring the part of e touching v, with colors also given by the shape of the edges (solid, dashed, …). We recover the standard notion of edge-coloring, which maps edges to colors, when for every edge e, $\mathrm{c}(e, _)$ has the same value for all endpoints of e. An example of locally colored graph is given on Figure 3, where $\mathrm{c}(e, v) = \mathrm{c}(e, u) = \mathrm{c}(f, u) = \mathrm{solid}$, $\mathrm{c}(f, w) = \mathrm{c}(g, v) = \mathrm{dashed}$, $\mathrm{c}(g, w) = \mathrm{solid}$ and $\mathrm{c}(h, v) = \mathrm{dotted}$.

A cusp at v of color α is a triple (e, v, f) where e and f are distinct edges such that v is an endpoint of both of these edges and $\mathrm{c}(e, v) = \alpha = \mathrm{c}(f, v)$. In this case, v is called the vertex of the cusp, α the color of the cusp and (v, α) is called a cusp-point. More generally, we will consider in this paper vertex-color pairs which are pairs made of a vertex and a color, and a cusp-point is a particular instance of a vertex-color pair. The locally colored graph in Figure 3 has two cusps, (e, u, f) and (f, u, e), both of vertex u and color solid, so that (u, solid) is the only cusp-point of this graph.

A cusp of a path p is a cusp made by a sub-sequence (e, v, f) of this path or, in case p is closed, a cusp (e_n, v_0, e_1) made by its last edge e_n, its source (and target) v_0 and its first edge e_1. Remark that the reverse of a path contains the same number of cusps as this path. A cusp-free path, also called an alternating path, is one without cusp. Given a non-empty path p, whose source is v_0 and first edge is e_1, its starting color is $\mathrm{c}(e_1, v_0)$. Similarly, if its target is v_n and its last edge is e_n, then the ending color of p is $\mathrm{c}(e_n, v_n)$. Remark the starting (resp. ending) color of $\overline{p}$ is the ending (resp. starting) color of p. For instance, in the graph depicted on Figure 3 the path (v, e, u, f, w) has one cusp at u of color solid, its starting color is solid and its ending one is dashed.

Fact 1.

Let ω be a cycle with no cusp at its source, and α a color. Then α is not the starting color of ω or α is not the starting color of $\overline{\omega}$.

We call splitting a vertex v such that any cycle containing it has a cusp at v. We will show in Section 6.1 that this fits the notion at play in the conclusion of Yeo's theorem [39].

▶ **Remark 2.**

We invented this "local coloring", which is not standard in the literature, and the name "cusp". When used only through the notions of cusps and splitting vertices, that a given color is used on different vertices has no impact. Hence, we could use different sets of colors depending on each vertex, or not use more colors than the maximal degree of the graph. Equivalently, a local coloring is an equivalence

relation on the edges incident to v, for each vertex v. We keep the idea of local coloring as it is a direct generalization of edge-coloring.

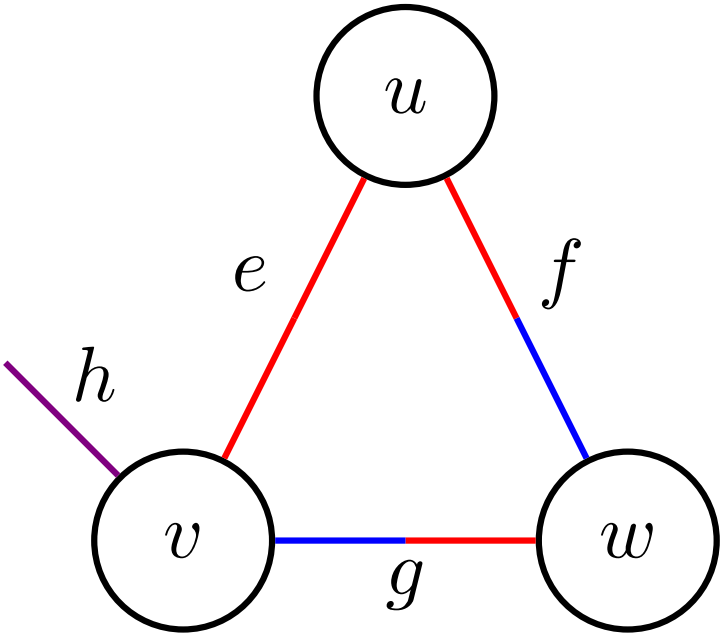


Figure 2: Example of locally colored graph.

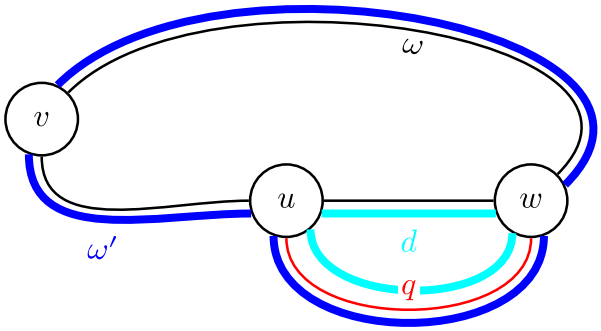


Figure 3: Illustration of Lemma 6.

3 A Generalization of Yeo's Theorem

We prove a version of Yeo's theorem [39] for locally colored partial graphs, which is moreover parametrized by the choice of a set of vertex-color pairs (subject to a technical condition): Theorem 8. We first fix a partial graph G with a local coloring c.

The main idea is to follow a path that is an evidence of progression, i.e. a strict partial order $\vec{\prec}$: a vertex is smaller than another when there is a path from the first to the second, and we will prove that a maximal vertex is splitting. As the hypothesis of the theorem is about cusp-free cycles, it makes sense to consider cusp-free paths in this ordering. However, two issues prevent $\vec{\prec}$ from being an order. First, the concatenation of two cusp-free paths may not be cusp-free. To have $\vec{\prec}$ transitive, we impose a condition on the starting and ending colors of the cusp-free path – which is why we consider vertex-color pairs and not simply vertices. Second, there is no reason for this relation $\vec{\prec}$ of "being linked by a cusp-free path" to not

loop. Hence, we add a condition on the path that there is no way to go back on it, yielding from $\overset{p}{\curvearrowright}$ a relation $\lhd$ which will be our strict partial order. This entails the following:

Definition 3.

Given vertices v and u, and colors α and β, we write $(v,\alpha) \overset{p}{\curvearrowright} (u,\beta)$ if p is a simple open cusp-free path from v to u with starting color not α and with ending color β. We note $(v,\alpha) \overset{p}{\lhd} (u,\beta)$ when $(v,\alpha) \overset{p}{\curvearrowright} (u,\beta)$ and for all vertex w, color τ and path q such that $(u,\beta) \overset{q}{\curvearrowright} (w,\tau)$, w is not in p. We simply write $(v,\alpha) \lhd (u,\beta)$ when such a path exists.

Lemma 4.

Let v, u and w be vertices, α, β and τ be colors, and p and q be paths. If $(v,\alpha) \overset{p}{\lhd} (u,\beta)$ and $(u,\beta) \overset{q}{\curvearrowright} (w,\tau)$ then $(v,\alpha) \overset{p\cdot q}{\curvearrowright} (w,\tau)$.

Lemma 5.

The relation $\lhd$ is a strict partial order on vertex-color pairs.

The key ingredient for proving our Yeo-style theorem is showing that for any pair (v,α) maximal for the strict partial order $\lhd$, v is splitting. It is a consequence of the following:

Lemma 6 (Cusp Minimization).

Fix a partial graph G with a local coloring. Assume ω is a cycle starting from a vertex v, with no cusp at v but containing a cusp of vertex u and color α. If $(u,\alpha) \overset{q}{\curvearrowright} (w,\beta)$ with w a vertex of ω, then either there exists a cusp-free cycle or there exists a cycle ω' starting from v, with no cusp at v and strictly less cusps than ω.

Proof.

Use Figure 3 as a reference for notations. W.l.o.g. q has no vertex in common with ω other than its endpoints u and w. We use the notation v_1 for the occurrence of v at the source of ω, and v_2 for its occurrence at the target of ω.

By symmetry (considering the reverse of ω if necessary), we can assume that w is in $\omega_{(u,v_2)}$ and if $w = v_2$ then β is not the starting color of ω. Indeed, if $w \notin \omega_{(u,v_2)}$, we reverse ω. Otherwise and if $w = v_2$, we apply Fact 1 to ω and β to get that ω or $\overline{\omega}$ respects our assumption.

Consider the cycles $\omega' = \omega_{(v_1,u)} \cdot q \cdot \omega_{(w,v_2)}$ and $d = q \cdot \overline{\omega}_{(w,u)}$ (see Figure 3). We count the cusps in ω, ω' and d. Recall that u is a cusp of ω of color α, q is cusp-free and its starting color is not α, and that ω' has no cusp at v (by our symmetry argument above). Thus, there are $n_1 + 1 + n_2 + b^\omega_w + n_3$ cusps in ω, $n_1 + b^{\omega'}_w + n_3$ in ω', and $b^d_w + n_2$ in d, where:

- n_1 (resp. n_2, n_3) is the number of cusps of $\omega_{(v_1,u)}$ (resp. $\omega_{(u,w)}$, $\omega_{(w,v_2)}$);
- b^ω_w (resp. $b^{\omega'}_w$, b^d_w) is 1 if ω (resp. ω', d) has a cusp at w and 0 otherwise.

If ω' has strictly less cusps than ω we are done, otherwise $b^{\omega'}_w \geq 1 + n_2 + b^\omega_w$. Hence, $n_2 = 0$, $b^\omega_w = 0$ and $b^{\omega'}_w = 1$. But the latter two imply $b^d_w = 0$, so that d is a cusp-free cycle. ◀

Proposition 7.

Let v be a non-splitting vertex of a locally colored partial graph which has no cusp-free cycle. For any color α there exists a cusp-point (u,β) such that $(v,\alpha) \lhd (u,\beta)$.

Proof.

Since v is not splitting, it is the source (and target) of a cycle ω which has no cusp at v. W.l.o.g. ω has a minimal number of cusps among all such cycles, and has not α as a starting color (thanks to Fact 1 and as reversing a path preserves the number and vertices of cusps). For ω cannot be cusp-free, it contains at least one cusp: denote by u the vertex of the first cusp of ω, and by β its color. We have $(v,\alpha) \overset{\omega_{(v,u)}}{\vec{\Gamma}} (u,\beta)$, and conclude $(v,\alpha) \overset{\omega_{(v,u)}}{\lhd} (u,\beta)$ by Lemma 6, the minimal number of cusps of ω and the absence of cusp-free cycles. ◀

A set P of vertex-color pairs dominates cusp-points if for any cusp-point (v,α), either $(v,\alpha) \in P$ or there is $(u,\beta) \in P$ with $(v,\alpha) \lhd (u,\beta)$. Our main result follows by Proposition 7.

Theorem 8 (Parametrized Local Yeo).
Take G a partial graph with a local coloring and pose P a set of vertex-color pairs which dominates cusp-points. If G has no cusp-free cycle, the vertex of any $\lhd$-maximal element of P is splitting.

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